When Physiological Plausibility Misleads: Covariance-Aligned Quality Gates for Synthetic Motor-Imagery EEG

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***Abstract*—Generative models are increasingly used to augment scarce motor-imagery (MI) electroencephalography (EEG) datasets, and their synthetic samples are commonly filtered by *physiological* quality gates such as the event-related desynchronization (ERD) that a motor-imagery trial is expected to produce. We show that this premise is systematically flawed: an ERD gate measures a *temporal* energy contrast, whereas the downstream common-spatial-pattern (CSP) classifier consumes the *spatial* covariance of each trial. Because the two criteria occupy orthogonal representational spaces, an ERD gate can be both blind to harmful corruption and itself damaging. We propose a dual gate that operates in the exact covariance feature space of the downstream decoder: a plausibility gate based on the cosine similarity of a trial's covariance to its class template, and a discriminative gate based on the Fisher linear-discriminant projection gap. On a real state-of-the-art generative model—DDF-Net, a denoising-diffusion model whose per-channel synthesis collapses cross-channel covariance (synthetic cosine similarity 0.368 versus real 0.996)—unfiltered samples degrade CSP accuracy from 0.833 to 0.667, whereas our dual gate admits $0/144$ samples (0% leakage) and losslessly restores the 0.833 baseline. These results argue that generative EEG models must jointly model the channel distribution, and that a quality gate is only effective when it shares the decoder's representation.**

***Index Terms*—EEG, motor imagery, synthetic data, generative models, diffusion, common spatial patterns, data quality.**

I. Introduction

Decoding motor imagery from EEG underpins non-invasive brain–computer interfaces (BCIs), but the scarcity and high inter-subject variability of labeled trials limit the performance of data-hungry decoders [1]–[4]. To relieve this bottleneck, researchers increasingly synthesize artificial MI-EEG with generative models—generative adversarial networks (GANs), variational autoencoders (VAEs), and, most recently, denoising diffusion probabilistic models (DDPMs)—and use the samples to augment training [5]–[14].

Because generated samples can be poor, a natural practice is to gate them by a *physiological* criterion before mixing them into the training set. In MI the dominant such criterion is the event-related desynchronization (ERD): a real motor-imagery trial should exhibit a band-power decrease over the contralateral sensorimotor cortex during the task [15]. This paper interrogates the implicit assumption behind that practice—that a *physiologically plausible* sample is necessarily a *useful* sample for the downstream decoder—and shows that the assumption does not hold.

The root cause is a representational mismatch. ERD is a *temporal* quantity: a ratio of band power between a baseline and a task window. The dominant MI decoder, CSP followed by a linear discriminant [1], [16], [17], consumes a *spatial* quantity: the per-trial channel covariance matrix $\mathbf{X}\mathbf{X}^\top$. A gate built on the former is therefore, in principle, blind to exactly the structure the latter exploits, and vice versa. We demonstrate three concrete consequences on controlled corruption: an ERD gate cannot detect white-noise corruption that destroys the covariance off-diagonal, it spuriously deletes harmless time-shifted trials, and in a full-data regime it can itself reduce accuracy. A covariance-aligned gate, in contrast, rejects the corruption while preserving the harmless perturbation.

Our contributions are threefold:

1) We identify and quantify the negative result that a physiological ERD gate is misaligned with the covariance representation of the CSP decoder, and can be both ineffective and harmful (Section II, Section IV).
2) We propose a *dual covariance-aligned gate* operating on the exact feature space of the downstream decoder: a plausibility gate using cosine similarity to the class covariance template, and a discriminative gate using the Fisher linear-discriminant projection gap (Section II).
3) We validate the gate against a real state-of-the-art generative model—DDF-Net, a diffusion-based MI generator—and show that its per-channel synthesis collapses spatial covariance, that unfiltered samples are actively harmful, and that our gate achieves 0% leakage while losslessly restoring baseline accuracy (Section IV, Section V).

II. Method

A. Problem setting and downstream decoder

We consider labeled MI-EEG trials $(\mathbf{X}_i, y_i)$, where $\mathbf{X}_i \in \mathbb{R}^{C\times T}$ is a trial with $C = 22$ EEG channels and T time samples, and $y_i \in \{-1, +1\}$ denotes left- versus right-hand imagery. The downstream decoder is a two-stage CSP + linear-discriminant-analysis (LDA) pipeline [1], [16]. CSP computes spatial filters from the per-class mean *spatial covariance*

matrices; that is, the decoder's only contact with a trial is through

$$\mathbf{C}_i \;=\; \mathbf{X}_i\,\mathbf{X}_i^{\top} \;\in \mathbb{R}^{C\times C}, \tag{1}$$

the uncentered energy covariance. Any quality gate that is to remain faithful to the decoder must therefore judge trials in the same covariance space.

We reduce $\mathbf{C}_i$ to a scale-invariant feature vector by trace normalization followed by vectorization of the upper triangle (including the diagonal),

$$\boldsymbol{\phi}_i \;=\; \mathrm{triu}\Big(\frac{\mathbf{C}_i}{\mathrm{tr}(\mathbf{C}_i)}\Big) \;\in\; \mathbb{R}^d, \qquad d = C(C+1)/2 = 253, \tag{2}$$

which matches the affine-invariance step of Riemannian covariance alignment and isolates cross-channel structure from overall trial power.

B. Gate 1: covariance plausibility

The first gate tests whether a trial *looks like* plausible EEG of its class. From the real calibration set we form a per-class covariance template $\boldsymbol{\mu}_c = \mathbb{E}[\boldsymbol{\phi} \mid y = c]$ and score each trial by the cosine similarity of its covariance features to its own class template,

$$\mathrm{sim}_{\cos}(\boldsymbol{\phi}, \boldsymbol{\mu}_c) \;=\; \frac{\boldsymbol{\phi}\cdot\boldsymbol{\mu}_c}{\|\boldsymbol{\phi}\|\,\|\boldsymbol{\mu}_c\|}. \tag{3}$$

An acceptance window is calibrated as the 5th–95th percentile of the within-class similarity distribution of real trials, $\mathcal{A}_c = \big[Q_5(\{s_i\}_{y_i=c}),\; Q_{95}(\{s_i\}_{y_i=c})\big]$, and a trial is retained iff $\mathrm{sim}_{\cos}(\boldsymbol{\phi}_i, \boldsymbol{\mu}_{y_i}) \in \mathcal{A}_{y_i}$. Because white-noise corruption inflates the diagonal (self-power) relative to the off-diagonal (cross-channel) entries, it rotates $\boldsymbol{\phi}$ away from $\boldsymbol{\mu}_c$ and drives the similarity out of the window. Crucially, however, a class-ambiguous blend of a left and a right trial is still a *valid* covariance and passes Gate 1; this motivates a second, discriminative gate.

C. Gate 2: discriminative projection gap

The second gate tests whether a trial carries the *correct class* information. We fit a Fisher LDA on the covariance features of real trials, obtaining a discriminative direction $\mathbf{w} \in \mathbb{R}^d$ (with a small ridge regularization) that best separates the two classes [18]. A trial is projected to a scalar score

$$z_i \;=\; \boldsymbol{\phi}_i^{\top}\mathbf{w}, \tag{4}$$

and per-class acceptance windows $\mathcal{D}_c = \big[Q_5(\{z_j\}_{y_j=c}),\, Q_{95}(\{z_j\}_{y_j=c})\big]$ are calibrated on real trials. A trial is retained iff $z_i \in \mathcal{D}_{y_i}$. For real MI data the two class score distributions are well separated with an intermediate *rejection gap*: a class-mixed trial, being an average of the two class covariances, projects near the midpoint between the class modes and lands in the gap, where it is rejected by both windows. The dual gate admits a trial only if it passes Gate 1 *and* Gate 2.

III. Experiments

A. Dataset and protocol

We use the BCI Competition IV dataset 2a, which records 22 EEG channels at 250 Hz from 9 subjects (A01–A09) performing four-class MI [19], [20]. We restrict to the left- and right-hand classes. Gates are calibrated on subject A01's training session (A01T) with a fixed $108/36$ trial split, and downstream accuracy is evaluated over 10 seeds; synthetic batches contain 144 trials (72 per class, balanced).

B. Controlled corruption modes

To probe the gates under known, interpretable defects, we generate three synthetic-batch families from real A01T trials: **Mode 1** (class mixing), averaging one left and one right trial, producing a valid but class-ambiguous covariance; **Mode 2** (white noise), adding 0 dB Gaussian noise, which attenuates cross-channel structure while inflating the diagonal; and **Mode 3** (time shift), a circular temporal shift, which preserves the covariance and is harmless to CSP.

C. Real generative model: DDF-Net

As a terminal test we use DDF-Net, a diffusion-based MI-EEG model that augments a dual-domain classification network with DDPM-generated samples [14]. We train its Stage-1 DDPM (a UNet-style architecture with $T = 100$ diffusion steps and 38.93 M parameters) on A01T and sample 144 synthetic trials. Because the generator models each of the channels *independently* and reassembles them into a trial afterwards, the joint cross-channel distribution is never learned—an ideal stress test for a covariance gate.

IV. Results

A. The gate is selective on controlled corruption

The dual gate behaves exactly as designed. On the A01T synthetic batch, Gate 1 rejects $0/144$ white-noise trials (Mode 2) but retains $123/144$ class-mixed trials (Mode 1), which are still valid covariances; Gate 2 then rejects the class-ambiguous blends, reducing Mode-1 retention to $50/144$. For time-shifted trials (Mode 3), whose covariance is unchanged, the dual gate correctly retains $114/144$, i.e., it does not spuriously kill a harmless perturbation. By contrast, an ERD gate retains $106/144$ of the white-noise trials while deleting roughly 20% of the harmless time-shifted trials.

Table I reports aggregate downstream accuracy across the 9 subjects at the most severe (100%) contamination level. On white noise (Mode 2), unfiltered augmentation drops accuracy from 0.762 to 0.690 (paired t-test $p < 0.001$), and the covariance-aligned gates restore it to 0.762, matching baseline. On class mixing (Mode 1) the dual gate recovers a modest but significant fraction of the loss at lower contamination ($p = 0.016$ at 25%, $p = 0.014$ at 50%); at 100% contamination the improvement is not significant ($p = 0.524$), consistent with a ceiling effect when abundant real data remain. Time shifts (Mode 3) induce no systematic degradation.

TABLE I

AGGREGATE DOWNSTREAM ACCURACY (MEAN ± STD OVER 9 SUBJECTS) AT 100% CONTAMINATION. MODE 1: CLASS MIXING; MODE 2: WHITE NOISE; MODE 3: TIME SHIFT. "COV" AND "DUAL" DENOTE THE COVARIANCE-ONLY GATE (GATE 1) AND THE DUAL GATE (GATES 1+2).

Mode	Base	Unf.	Cov	Dual
1 (mix)	0.762±0.125	0.732±0.117	0.737±0.112	0.739±0.128
2 (noise)	0.762±0.125	0.690±0.156	0.762±0.125	0.762±0.125
3 (shift)	0.762±0.125	0.740±0.126	0.743±0.128	0.744±0.130

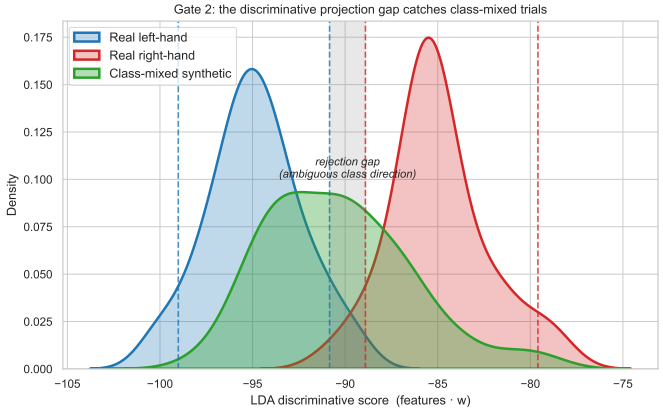


Fig. 1. Fisher LDA discriminative scores of real left- and right-hand trials (blue/red) and class-mixed synthetic trials (green). The two classes' 5%–95% acceptance windows are separated by a shaded *rejection gap*; class-ambiguous blends project near the midpoint and fall into this gap, where Gate 2 rejects them.

B. Real-model validation: DDF-Net collapses covariance

Against the real DDF-Net generator the gate is decisive. Fig. 2 visualizes the root cause: the trace-normalized mean covariance of DDF-Net synthetic right-hand trials is nearly diagonal, whereas real trials carry substantial off-diagonal (cross-channel) structure; the difference map isolates exactly the structure that per-channel generation discards. Fig. 3 translates this into the gate's decision variable: real trials concentrate at cosine similarity 0.996 to their class template, while synthetic trials cluster at 0.368—well below the $[0.990, 0.999]$ acceptance window, so $144/144$ synthetic trials fall outside and are rejected (0% leakage).

Table II reports the terminal result. Feeding all 144 unfiltered synthetic trials into training degrades CSP accuracy from 0.8333 to 0.6667, a 0.17 drop that confirms the synthetic data are not merely uninformative but actively harmful. Applying the covariance gate (or the full dual gate) keeps $0/144$ samples and restores accuracy to the 0.8333 baseline with no loss.

TABLE II

DOWNSTREAM CSP ACCURACY ON DDF-NET SYNTHETIC AUGMENTATION. "KEPT" IS THE NUMBER OF SYNTHETIC TRIALS ADMITTED INTO TRAINING.

Condition	Accuracy	Kept
Baseline (no augmentation)	0.8333	–
Unfiltered	0.6667	144/144
Covariance gate (Gate 1)	0.8333	0/144
Dual gate (Gates 1+2)	0.8333	0/144

V. DISCUSSION

A. Why per-channel generation is harmful

CSP is a spatial method: its filters are the eigenvectors of a generalized eigenvalue problem formed from per-class *covariance* matrices [1], [16]. A generator that synthesizes each channel independently and stitches channels together afterwards can reproduce each channel's marginal power spectrum, but it cannot reproduce the cross-channel correlations that constitute the spatial structure CSP exploits. The result is the diagonalization visible in Fig. 2: the synthetic trials occupy a region of covariance space so far from the real class templates (cosine similarity 0.368 versus 0.996) that even a coarse plausibility gate rejects them all. This is a stronger statement than "the samples are noisy" —it shows that a *per-channel* generative model is architecturally incapable of producing CSP-useful data, regardless of how realistic its single-channel signals appear. We therefore argue that generative EEG models intended for spatial decoders must model the *joint* channel distribution.

B. Why the representation must match the decoder

The negative result on the ERD gate follows the same logic in reverse. ERD is a temporal band-power contrast; it is orthogonal to the spatial covariance the decoder reads. Consequently an ERD gate is blind to covariance collapse and can simultaneously reject harmless temporal perturbations. Our dual gate sidesteps this by construction: both of its decision variables—cosine similarity and the LDA projection—are computed on the identical trace-normalized covariance features that the decoder consumes. The alignment, not the specific threshold, is what makes the gate effective, and we expect the principle to transfer to any decoder whose representation can be made explicit.

C. Limitations

Three limitations bound our conclusions. First, a ceiling effect: when real data are abundant (e.g., Mode 1 at 100% contamination), the marginal value of the gate disappears, so significance is expected to vanish. Second, boundary leakage: class-mixed discriminative scores follow a distribution centered near the class boundary, so a fraction of Mode-1 blends still land inside a class window ($50/144$ retained); this is the inherent cost of a percentile window, and tightening it would misclassify the harmless Mode-3 trials. Third, our gate is calibrated on a single subject and session (A01T); cross-subject and cross-session generalization remains to be established.

VI. CONCLUSION

We showed that physiological plausibility is an unreliable proxy for downstream utility in synthetic MI-EEG: an ERD gate operates in a temporal space orthogonal to the spatial

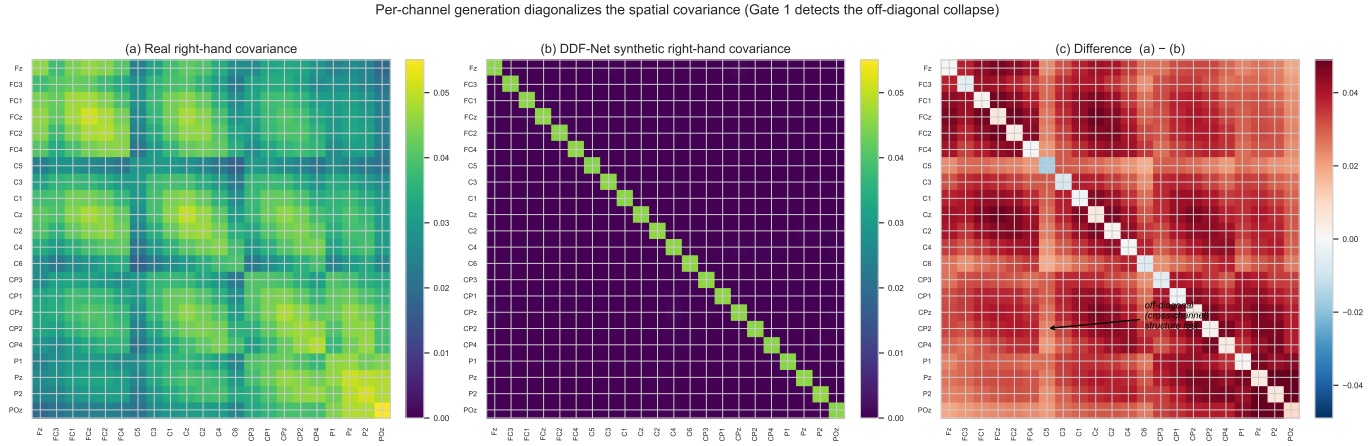


Fig. 2. Trace-normalized mean spatial covariance (22×22) of (a) real right-hand trials, (b) DDF-Net synthetic right-hand trials, and (c) their element-wise difference. Per-channel independent generation produces a near-diagonal synthetic covariance, discarding the off-diagonal (cross-channel) structure on which CSP spatial filters depend.

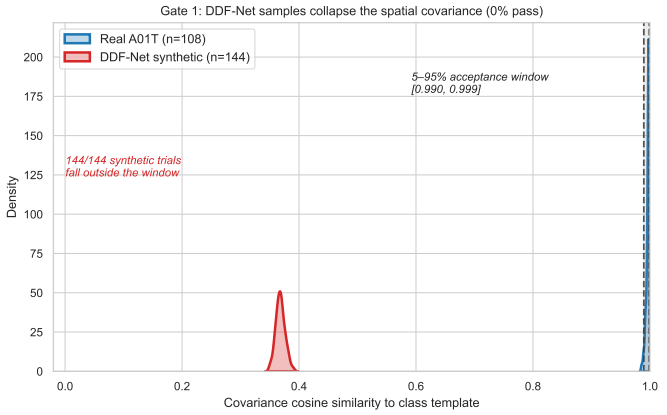


Fig. 3. Distributions of per-trial covariance cosine similarity to the class template for real A01T trials (blue) and the 144 DDF-Net synthetic trials (red), with the 5%–95% acceptance window shaded. Synthetic similarity (mean 0.368) collapses far below the window [0.990, 0.999]; all 144 trials fall outside, so Gate 1 rejects 100%.

covariance that CSP decodes, and can be both blind to harmful corruption and itself damaging. A dual gate aligned to the decoder's covariance representation—cosine-similarity plausibility plus a discriminative projection gap—recovers downstream accuracy under controlled corruption and, against the real DDF-Net diffusion generator, achieves 0% leakage (0/144) while losslessly restoring the 0.833 baseline. The finding that per-channel synthesis collapses spatial covariance carries a direct design imperative: generative EEG models for spatial decoders must model the joint channel distribution, and quality gates must share the decoder's representation to serve as the last line of defense before deployment.

References

[1] B. Blankertz, R. Tomioka, S. Lemm, M. Kawanabe, and K.-R. Müller, "Optimizing spatial filters for robust eeg single-trial analysis," *IEEE Signal Processing Magazine*, vol. 25, no. 1, pp. 41–56, 2008.

[2] R. T. Schirrmeister, J. T. Springenberg, L. D. J. Fiederer, M. Glasstetter, K. Eggensperger, M. Tangermann, F. Hutter, W. Burgard, and T. Ball, "Deep learning with convolutional neural networks for eeg decoding and visualization," *Human Brain Mapping*, vol. 38, no. 11, pp. 5391–5420, 2017.

[3] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C. P. Hung, and B. J. Lance, "Eegnet: a compact convolutional neural network for eeg-based brain–computer interfaces," *Journal of Neural Engineering*, vol. 15, no. 5, p. 056013, 2018.

[4] F. Fahimi, Z. Zhang, W. B. Goh, T.-S. Lee, K. K. Ang, and C. Guan, "Inter-subject transfer learning with an end-to-end deep convolutional neural network for eeg-based bci," *Journal of Neural Engineering*, vol. 16, no. 2, p. 026007, 2019.

[5] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, "Generative adversarial nets," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 27, 2014, pp. 2672–2680.

[6] D. P. Kingma and M. Welling, "Auto-encoding variational bayes," in *International Conference on Learning Representations (ICLR)*, 2014.

[7] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 6840–6851.

[8] P. Dhariwal and A. Nichol, "Diffusion models beat gans on image synthesis," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021, pp. 8780–8794.

[9] Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole, "Score-based generative modeling through stochastic differential equations," in *International Conference on Learning Representations (ICLR)*, 2021.

[10] K. G. Hartmann, R. T. Schirrmeister, and T. Ball, "Eeg-gan: Generative adversarial networks for electroencephalographic (eeg) brain signals," arXiv preprint arXiv:1806.01875, 2018.

[11] E. Lashgari, D. Liang, and U. Maoz, "Data augmentation for deep-learning-based electroencephalography," *Journal of Neuroscience Methods*, vol. 346, p. 108885, 2020.

[12] J. Vetter, J. H. Macke, and R. Gao, "Generating realistic neurophysiological time series with denoising diffusion probabilistic models," *Patterns*, vol. 5, no. 9, p. 101047, 2024.

[13] M. D. Dere, J.-H. Jo, and B. Lee, "Motor-intent decoding from synthetic eeg data using denoising diffusion probabilistic models," *Expert Systems with Applications*, vol. 299, p. 130134, 2025.

[14] X. Li, J.-H. Wang, J. An, L.-L. Ren, L.-F. Bian, and C. Yang, "Diffusion enhancement-based dual-domain fusion attention motor imagery electroencephalography classification network," *Engineering Applications of Artificial Intelligence*, vol. 167, p. 113954, 2026.

[15] G. Pfurtscheller and F. H. Lopes da Silva, "Event-related eeg/meg synchronization and desynchronization: basic principles," *Clinical Neurophysiology*, vol. 110, no. 11, pp. 1842–1857, 1999.

[16] H. Ramoser, J. Müller-Gerking, and G. Pfurtscheller, "Optimal spatial filtering of single trial eeg during imagined hand movement," *IEEE Transactions on Rehabilitation Engineering*, vol. 8, no. 4, pp. 441–446, 2000.

[17] F. Lotte, L. Bougrain, A. Cichocki, M. Clerc, M. Congedo, A. Rakotomamonjy, and F. Yger, "A review of classification algorithms for eeg-based brain–computer interfaces: a 10 year update," *Journal of Neural Engineering*, vol. 15, no. 3, p. 031005, 2018.

[18] R. A. Fisher, "The use of multiple measurements in taxonomic problems," *Annals of Eugenics*, vol. 7, no. 2, pp. 179–188, 1936.

[19] M. Tangermann, K.-R. Müller, A. Aertsen, N. Birbaumer, C. Braun, C. Brunner, R. Leeb, C. Mehring, K. J. Miller, G. R. Müller-Putz *et al.*, "Review of the bci competition iv," *Frontiers in Neuroscience*, vol. 6, p. 55, 2012.

[20] C. Brunner, R. Leeb, G. Müller-Putz, A. Schlögl, and G. Pfurtscheller, "Bci competition 2008–graz data set a," Institute for Knowledge Discovery (Laboratory of Brain-Computer Interfaces), Graz University of Technology, Tech. Rep., 2008.